

JKBOSE 2026

CLASS 11TH

Morphology of Flowering Plants

LECTURE-01

Biology

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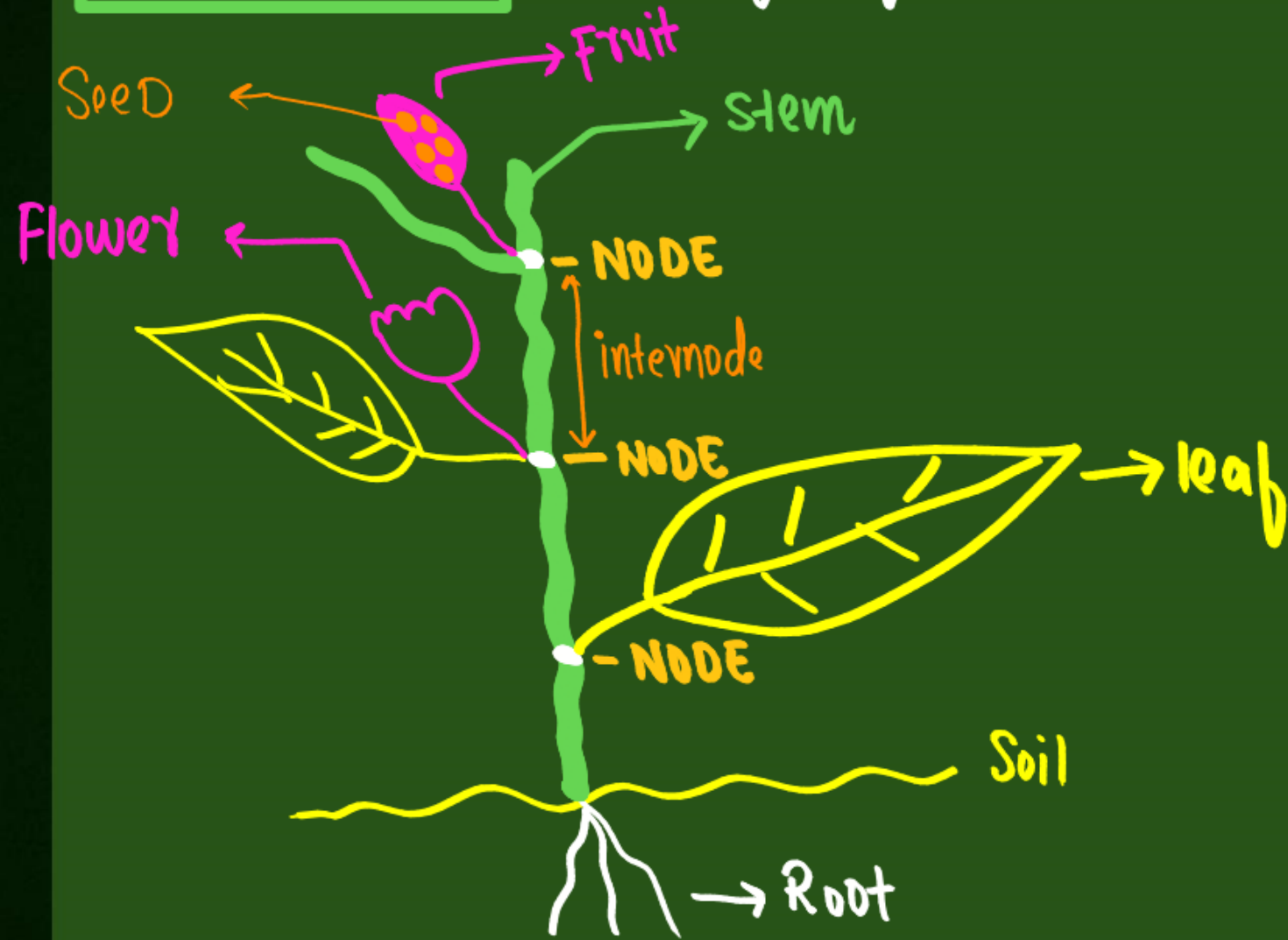


Topics To Covered

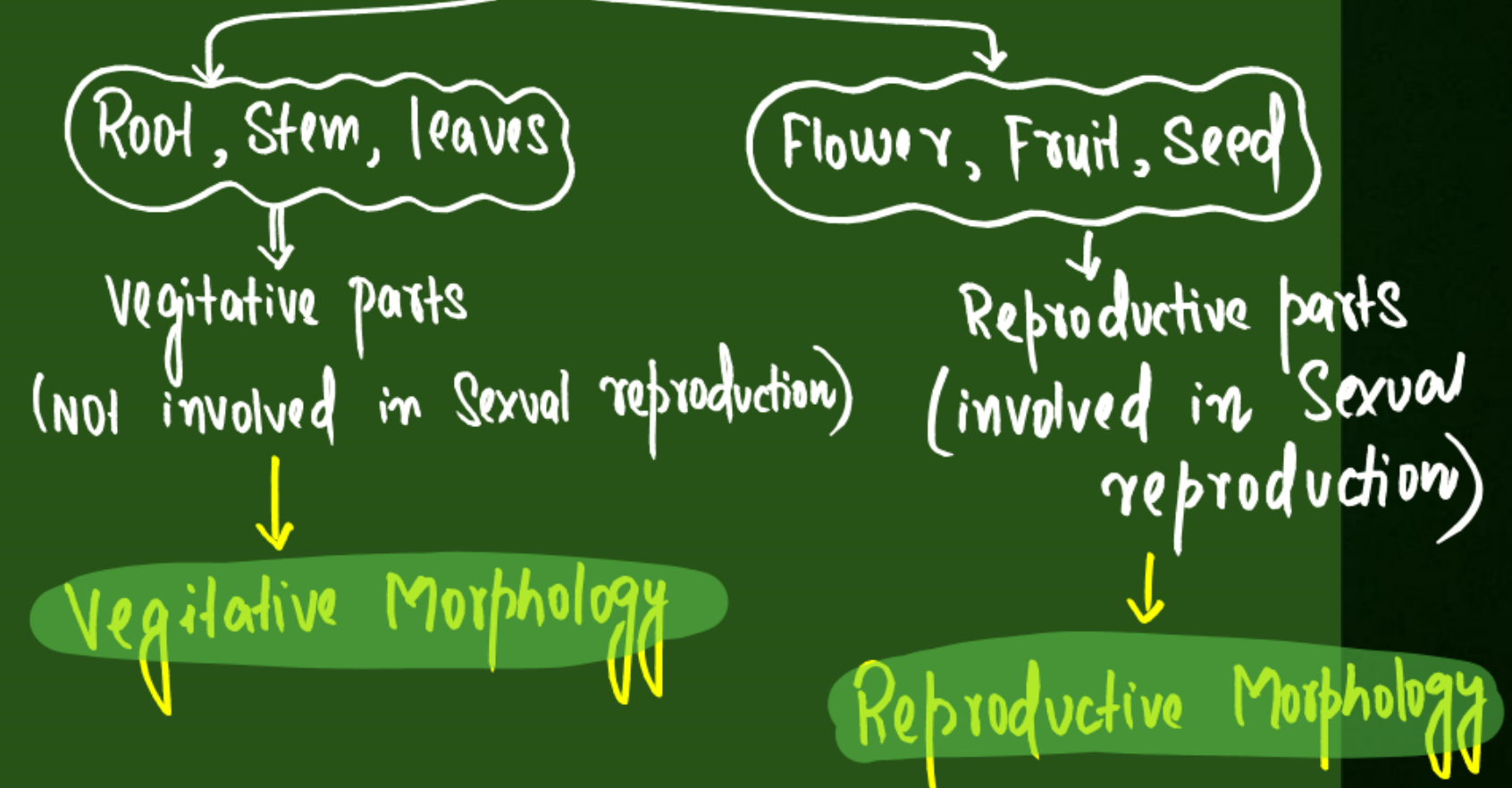
- 1 INTRODUCTION
- 2 MORPHOLOGY OF ROOT
- 3 BASIC FEATURES OF ROOT
- 4 REGIONS OF ROOT
- 5 MAIN FUNCTIONS OF ROOT
- 6 MODIFICATIONS OF ROOT → Not in JKBOSE syllabus.
→ Not in NEET syllabus.

INTRODUCTION

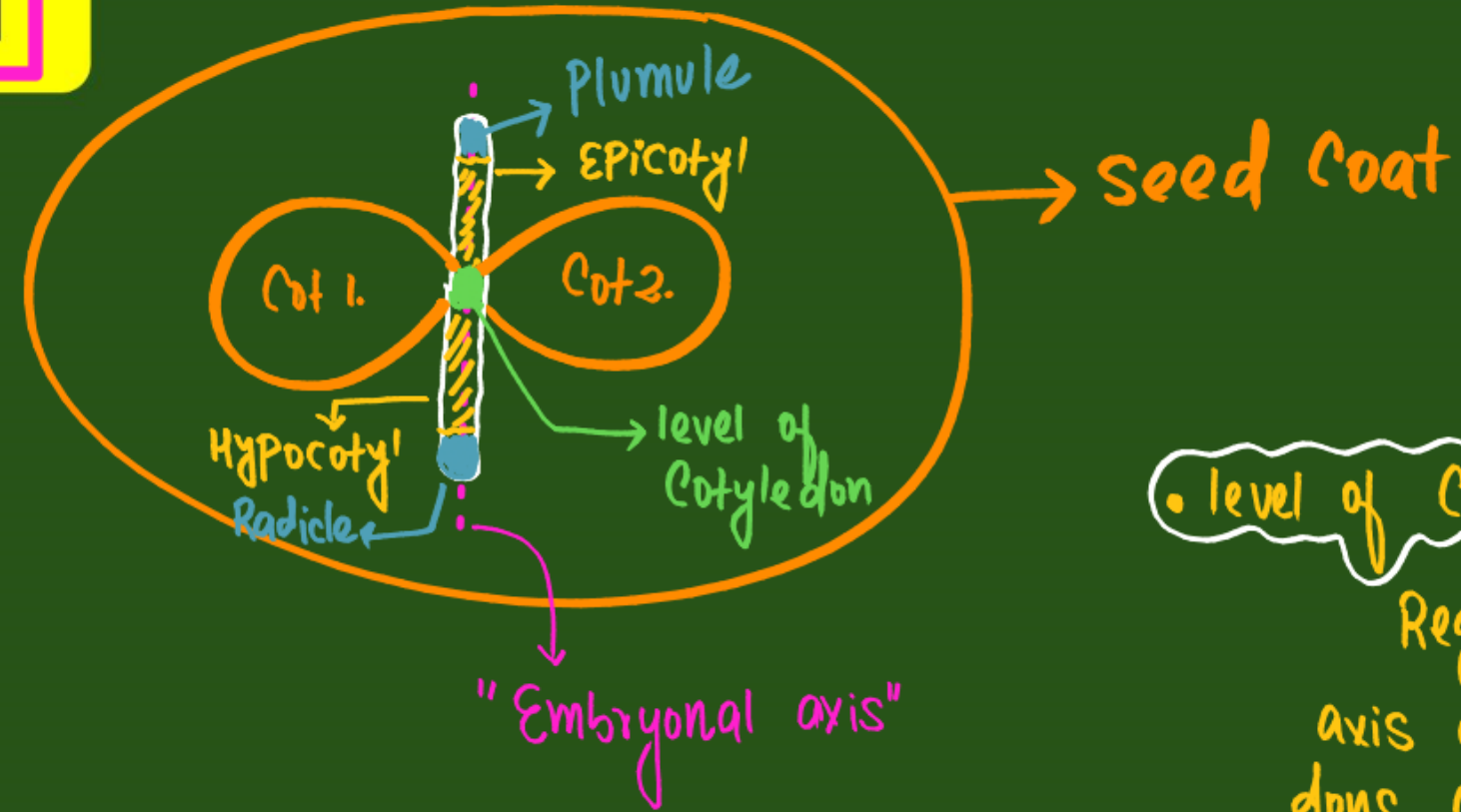
• **Morphology** — Study of External characters of an organism.



* Angiosperm (Flowering plant)



MORPHOLOGY OF ROOT



• level of Cotyledon → Region of Embryonal axis at which two Cotyledons are attached.

• Epicotyl → Region of Embryonal axis above level of Cotyledon

• Plumule → tip of Epicotyl.
→ Forms Shoot of plant.

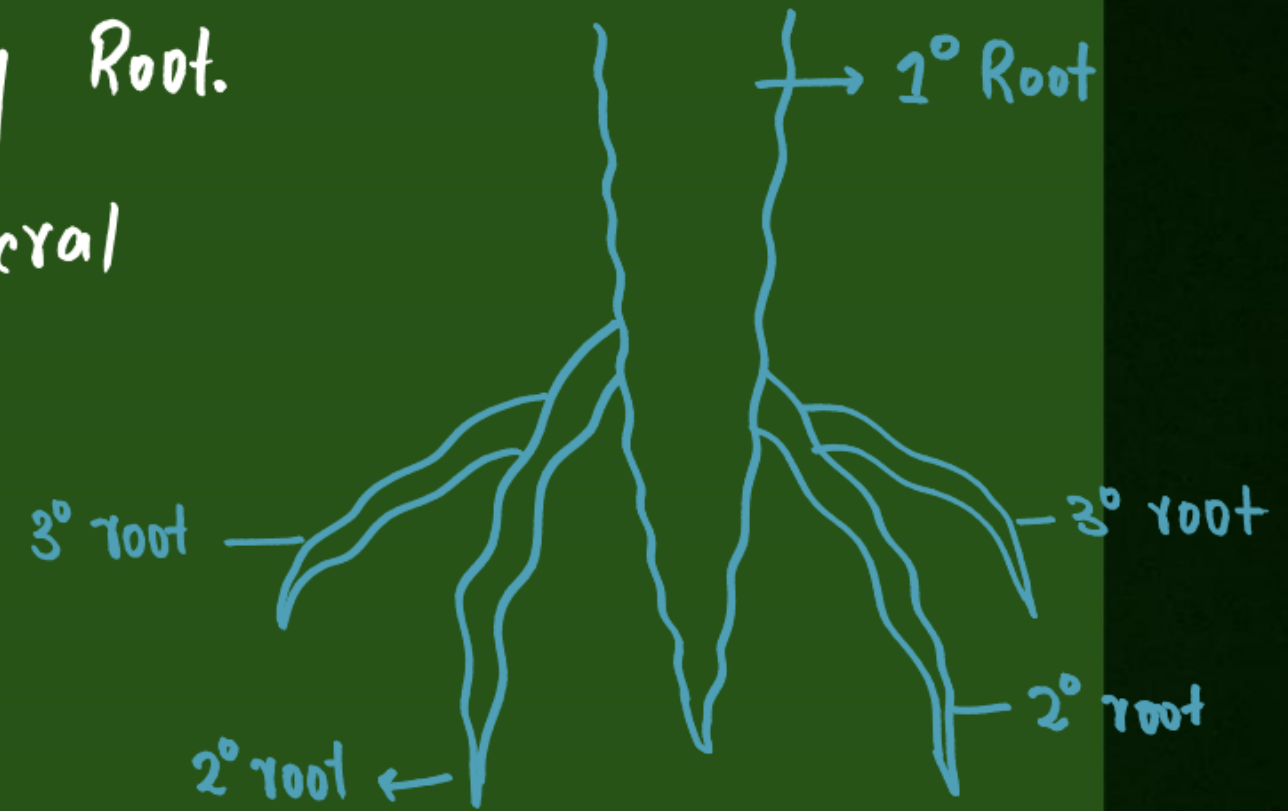
• Hypocotyl → Region of Embryonal axis below level of Cotyledon.

• Radicle → tip of hypocotyl.
→ Forms root of Plant.

Dicot Root

- Direct Elongation of Radicle forms Primary Root.
- Primary root bears lateral branches of several orders like 2° root, 3° root etc.
- Primary root along with its lateral branches is as **tap root**.

Ex. Mustard.



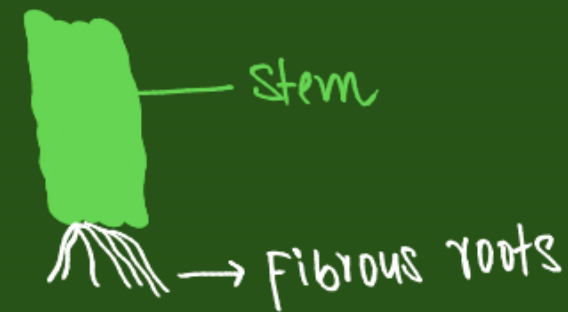
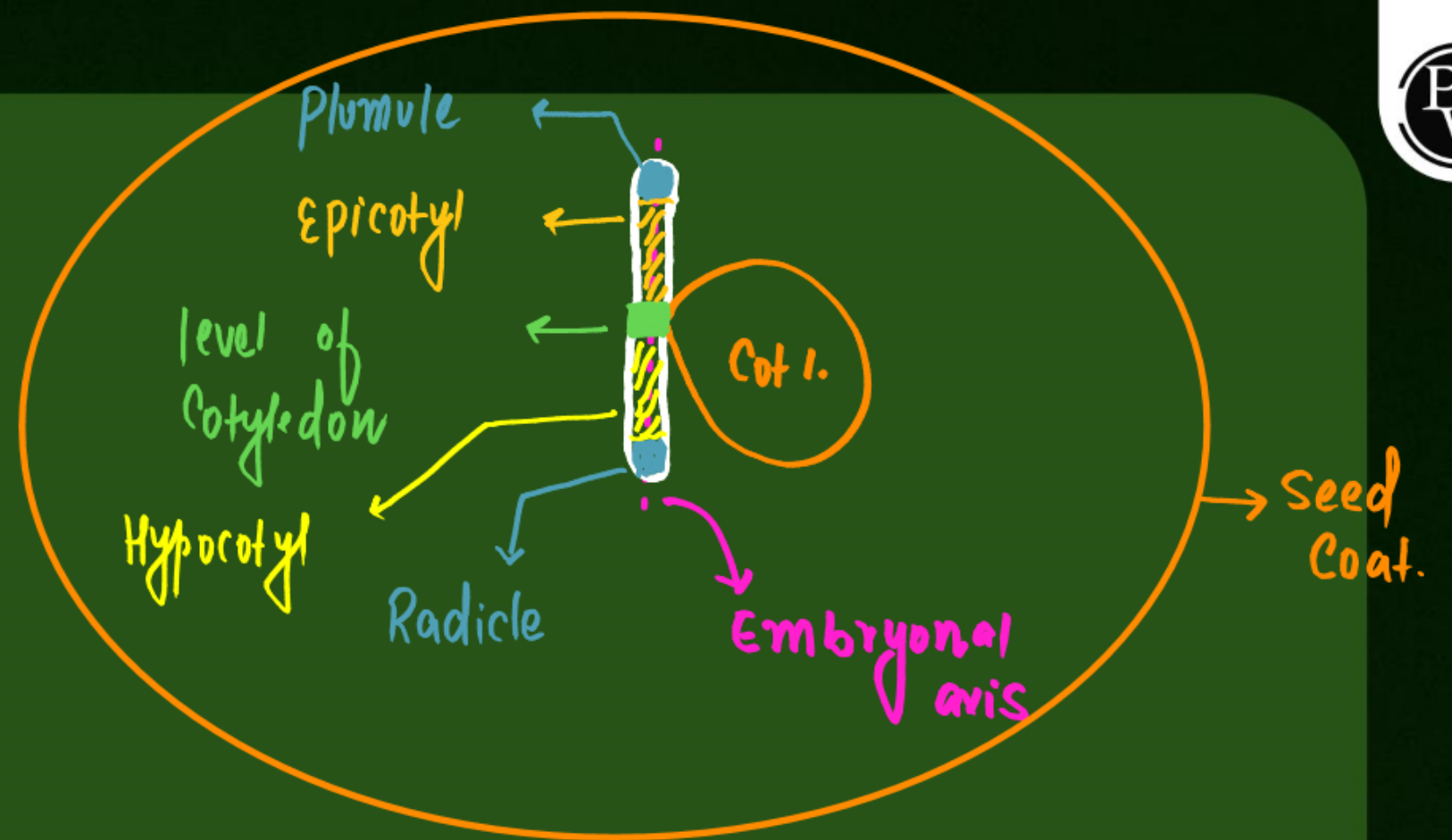
MONOCOT ROOT

- In Monocots, Radicle directly elongates and forms primary Root.
- In Monocots, primary root is short lived.
- Here, primary root is replaced by

Fibrous Roots.

- Originates from Base of Stem.
- Numerous.
- Oblique
- It is a type of Adventitious Root.

Example: wheat, Barley.



ADVENTITIOUS ROOT

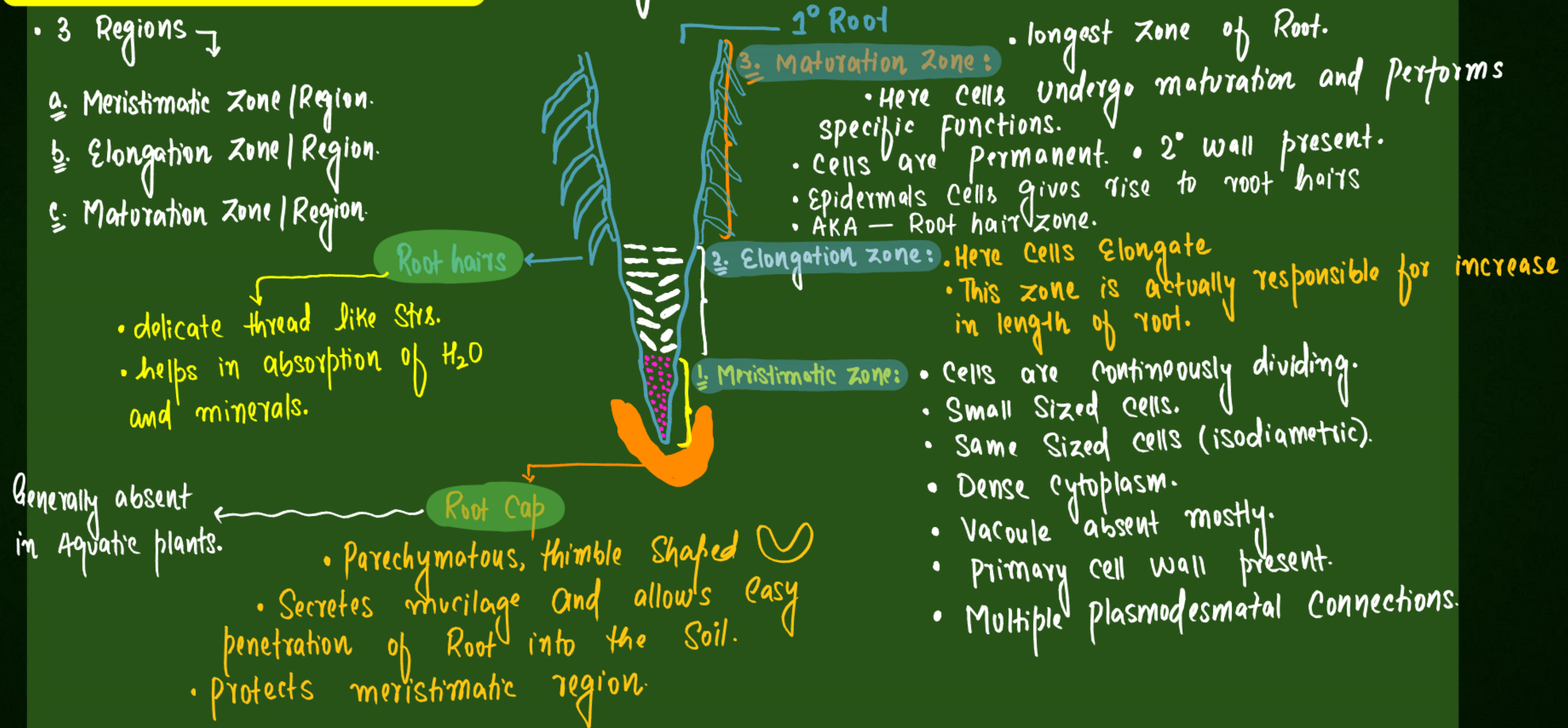
- underground / aerial.
- May be present in Monot as well as Dicots.
- It arises from part of plant other than radicle.
- Example : Fibrous roots.
Banyan tree (Ficus bengalensis)
Monstera

REGIONS OF ROOT

→ Primary Root.

• 3 Regions →

- Meristematic Zone / Region.
- Elongation Zone / Region.
- Maturation Zone / Region.



MAIN FUNCTIONS OF ROOT

↓
Next lecture

MODIFICATIONS OF ROOT

↓
Nex lecture.

QUESTIONS

Root hairs develop from the region of:

(2017-Delhi)

- ☒ **A** Maturation
- ☐ **B** Elongation
- ☐ **C** Root cap
- ☐ **D** Meristematic activity

QUESTIONS

u/w

In a longitudinal section of a root, starting from the tip upward, the four zones occur in the following order: (2004)

- A** Root cap, cell division, cell maturation, cell enlargement
- B** Cell division, cell enlargement, cell maturation, root cap
- C** Cell division, cell maturation cell enlargement, root cap
- D** Root cap, cell division, cell enlargement, cell maturation

QUESTIONS

H/w

The roots that originate from the base of the stem are:

[NEET 2020]

- A** Lateral roots
- B** Fibrous roots
- C** Primary roots
- D** Prop roots

Home work

2 Questions



Thank
You